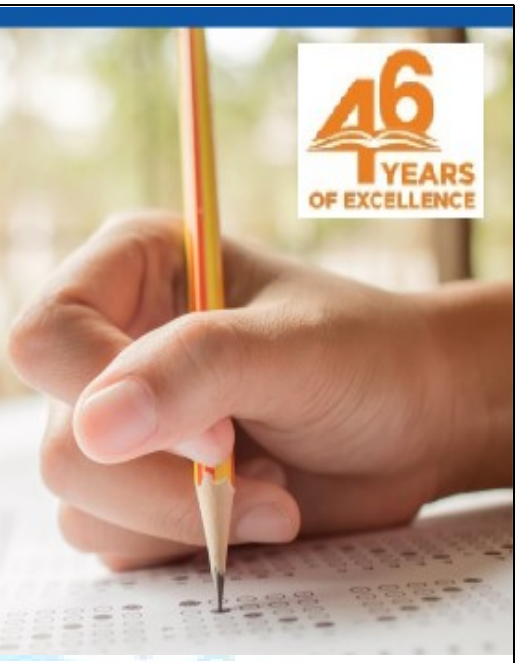
NARAYANA IIT ACADEMY INDIA

JEE ADVANCED GRAND TEST SERIES



46 YEARS OF EXCELLENCE

NARAYANA IIT - JEE ACADEMY - INDIA

Sec: OSR.IIT _*CO-SC(MODEL-A&APEX) **Date: 22-03-26**
Time: 3 Hrs **Max. Marks: 198**

22-03-26_OSR.IIT_*CO-SC(MODEL-A&APEX)_JEE ADV_GTA-1(P1)_SYLLABUS

Rankers Academy JEE

MATHEMATICS: **TOTAL SYLLABUS**

PHYSICS: **TOTAL SYLLABUS**

CHEMISTRY: **TOTAL SYLLABUS**

THE NARAYANA GROUP

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TIME: 3HRS **IMPORTANT INSTRUCTIONS** Max Marks: 198

MATHEMATICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 6)	Questions with Single Correct Options	+3	-1	6	18
Sec – II (Q.N : 7 – 12)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-2	6	24
Sec – III (Q.N : 13 – 18)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -1 27.30)	+4	0	6	24
Total				**18**	**66**

PHYSICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 19 – 24)	Questions with Single Correct Options	+3	-1	6	18
Sec – II (Q.N : 25 – 30)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-2	6	24
Sec – III (Q.N : 31 – 36)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)	+4	0	6	24
Total				**18**	**66**

CHEMISTRY

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 37 – 42)	Questions with Single Correct Options	+3	-1	6	18
Sec – II (Q.N : 43 – 48)	One of More Correct Options Type (partial marking scheme) (+1)	+4	-2	6	24
Sec – III (Q.N : 49 – 54)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)	+4	0	6	24
Total				**18**	**66**

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MATHEMATICS **MAX.MARKS: 66**

SECTION – 1 (Maximum Marks: 18)

This section contains SIX (06) questions.
Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer.
For each question, choose the correct option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If only the correct option is chosen.
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)
Negative Marks: -1 In all other cases.

1. "Sudheer has 7 friends. He writes 7 letters, each uniquely meant for one friend, and places them randomly into 7 addressed envelopes, each bearing the specific name of the intended friend.

Out of the seven friends :

1) Two of them are brothers and live at the same address.

2) The other five friends live at five different addresses.

Find the number of ways Sudheer can place the letters into the addressed envelopes such that none of the friends receives the letter meant for them."

(Assume that letters received at the two brothers address are received by both)

A) 1280 B) 1368 C) 684 D) 640

2. If the area bounded by $x=\sqrt{r^2-y^2}$ and $|y|=|\tan\theta(x+r)|$ is $\frac{1}{2}\pi r^2$ then (where $r\in R^+$ and $0<\theta<\frac{\pi}{2}$)

A) $2\theta-\sin 2\theta=\frac{\pi}{2}$ B) $\sin\theta=\theta$ C) $2\theta+\sin 2\theta=\frac{\pi}{2}$ D) $\sin 2\theta+2\theta=\pi$

3. $\int_1^2 \frac{x(x+1)dx}{(2x^3+3x^2-1)(4x^6+12x^5+9x^4-6x^3-9x^2+2)^{\frac{1}{2}}}=$

A) $\frac{2\sqrt{78}+9\sqrt{3}}{54}$ B) $\frac{2\sqrt{78}-9\sqrt{3}}{54}$

C) $\frac{9\sqrt{3}-2\sqrt{78}}{54}$ D) $\frac{2\sqrt{26}-3\sqrt{3}}{18}$

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4. Let $f:R\to R$ be a differentiable function such that

$\int_0^x \left(t.f(x)-x.f(t)\right)t\,dt=0,\ \forall x\in R$ and $f(2)=\frac{-1}{3}$ then

A) f(0) = -1 B) f(0) = 1 C) f(3) = -2 D) f(6) = -1

5. Let $A=\begin{bmatrix}\cos\left(\frac{\pi}{5}\right) & -\sin\left(\frac{\pi}{5}\right)\\ \sin\left(\frac{\pi}{5}\right) & \cos\left(\frac{\pi}{5}\right)\end{bmatrix}$ and $B=\begin{bmatrix}0 & -1\\ 1 & 0\end{bmatrix}$

If $\sum_{r=1}^{4}\left(B^2-B.A^{9r}-A^r.B+A^{10r}\right)=\begin{bmatrix}a & b\\ c & d\end{bmatrix}$, then $a^{10}+b^{10}+c^{10}+d^{10}=$

A) 0 B) $2\left(\cos^{10}\left(\frac{\pi}{5}\right)+\sin^{10}\left(\frac{\pi}{5}\right)\right)$ C) 4 D) 2

6. Sudheer has 3 identical white cubes and wants to turn them into dice. For each of the three cubes, he will make two independent decisions :

1. The colour : He will use either a Red or a Blue marker.

2. The Pattern : He will use either Dots or Digits.

For each cube, he flips a fair coin to decide the colour and another fair coin to decide the pattern (resulting in 6 coin flips in total to create the set of 3 dice).

After making the dice, Sudheer places all 3 dice into a bag. Renuka then randomly selects one die from the bag and notes its colour and pattern.

Consider the following events :

a. Event A : The die Renuka selects is Red.

b. Event B : The die Renuka selects is Dotted.

The probability that the events A and B are independent is

A) $\frac{7}{16}$ B) $\frac{21}{64}$ C) $\frac{15}{32}$ D) 0

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SECTION - 2 (Maximum Marks : 24)

This section contains SIX (06) questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

7. Which of the following is/are odd functions ? (where [.] denotes G.I.F)

A) $f:R\to R$ such that $f(x)=\dfrac{|x|.x}{1+x+x^2}$

B) $f:Z\to R$ such that $f(x)=\dfrac{x.[x].\tan x}{x^4+x^2+1}$

C) $f:R-Q\to R$ such that $f(x)=\sin\left(\dfrac{\pi}{2}\left[x+\dfrac{1}{2}\right]\right)$

D) $f:\left\{4k\pm\dfrac{1}{2}:k\in Z\right\}\to R$ such that $f(x)=\sin\left(\dfrac{\pi}{2}\left[x+\dfrac{1}{2}\right]\right)-\dfrac{1}{2}$

8. Let $f:R^+\to R$ be a twice differentiable function such that $f''(x)>0$, for all $x>0$ and $f(x+1)-f(x)$ is a positive integer for all positive integral values of x.

If $f(2025)=f(2026)-2027$ and $f(1)=\dfrac{1}{2}$ then f(3) can be

A) 9.5 B) 8.5 C) 7.5 D) 6.5

9. Let M be a point on the ellipse $\dfrac{x^2}{2}+\dfrac{y^2}{1}=1$. Let normal at M meets the ellipse at N. Tangents drawn to the ellipse at M and N intersect at $S(\alpha,\beta)$ such that α and β are both integers. Then

A) $\dfrac{8}{\alpha^2}+\dfrac{1}{\beta^2}=1$ B) $\dfrac{3}{\alpha^2}+\dfrac{6}{\beta^2}=1$

C) Number of such possible points of S is 4 D) $\tan(\angle MNS)=\dfrac{3}{2}$

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10. If $f:[0,1]\to(0,\infty)$ is a continuous function such that $\int_0^1 f(x)dx=1$ then which of the following is/are always **TRUE**?

A) $\left(\int_0^1 \sqrt[3]{f(x)}\,dx\right).\left(\int_0^1 \sqrt[5]{f(x)}.dx\right)\le 1$ B) $\int_0^1 \frac{f(x)}{1+2f(x)}dx\le\frac{1}{3}$

C) $\left(\int_0^1 \sqrt[4]{f(x)}dx\right)^2\ge 1$ D) There exists $c\in(0,1)\ni f(c)=1$

11. Let S be the set of all points in three dimensional space whose position vector, $\vec{r}$ satisfies $\vec{r}.\hat{i}=0$ and $\left|\vec{r}.\left(\hat{j}-2\vec{k}\right)\right|=\left|\vec{r}.\left(2\hat{j}+\hat{k}\right)\right|$ then

A) Number of points in 'S', whose distance from origin is 1, is 4

B) Number of points in 'S', whose distance from origin is 1, is infinite

C) Let P, Q be two points which belongs to set S such that OP = OQ = $\sqrt{10}$ and O, P, Q are non collinear, where O is origin. Then mid point of P, Q can be (0, 1, -2)

D) Let P, Q be two points which belongs to set S such that OP = OQ = $\sqrt{10}$ and O, P, Q are non collinear, where O is origin. Then mid point of P, Q can be (0, 2, -1)

12. Let S be the set of all possible value of 'n' for which the following list of '3n' numbers has unique mode

1, 2, 3, …., n, $1^2,2^2,3^2,\ldots..,\ n^2,1^3,2^3,3^3,\ldots.,n^3$

Let 'k' be the median of the following list of 4052 numbers

1, 2, 3, ….,2026, $1^2,2^2,3^2,\ldots.,2026^2$

then which of the following options is/are **CORRECT**?

A) The sum of elements in the set S is equal to 2016

B) The sum of elements in the set S is greater than 2026

C) k = 1982.5

D) k = 1981.5

For More Join: @IITJEE_Advanced

SECTION - 3 (Maximum Marks : 24)

This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round- off** the value to **TWO** decimal places.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

13. Let $f:[1,\infty)\to[1,\infty)$ be a function such that

$$f(x)=\begin{cases}2n.\left(\dfrac{2n-1}{2n}\right)^{x-2n+1} & ;2n-1\le x\le 2n;\forall n\in N\\ (2n+1)\left(\dfrac{2n}{2n+1}\right)^{x-2n} & ;2n<x<2n+1;\forall n\in N\end{cases}$$

Then the number of values of 'x' in [1, 5] for which $f(x)=f^{-1}(x)$ is

14. Let $f(x)=\lim\limits_{n\to\infty}\left[\dfrac{12xn^2+84x^2-12x}{n^2+10x^2}\right]$; where [.] denotes G.I.F then number of positive integers, 'm' which satisfy the equation $f\left(\dfrac{1}{m}\right)-\dfrac{12}{m}=0$ is

15. Let z_1,z_2,z_3 are three distinct complex numbers such that $|z_1|=|z_2|=|z_3|=4$ and $\left|z_1.(\overline{z_2}-\overline{z_3})-\overline{z_1}.(z_2-z_3)+z_2\overline{z_3}-z_3\overline{z_2}\right|=48$ then

$\left|(z_1-z_2)(z_2-z_3)(z_3-z_1)\right|$ is equal to

16. Let $b>a>0$ and A_1,H_1 be arithmetic mean and Harmonic mean of 'a' and 'b' respectively. Let A_n,H_n be respectively arithmetic mean and Harmonic mean of 'a' and A_{n-1} for all $n\ge 2$, where $n\in N$. If $A_{2025}=H_{2024}$ then number of divisors of $\left(\dfrac{b-a}{a}\right)$ that are less than 2025 is

17. Let A, B be variable points on the parabola $y^2=12x$ and foot of perpendiculars drawn from A, B on to directrix of the parabola be N, K respectively. Let S be the focus of the parabola. Then the locus of circumcentre of ΔNSK, as A, B varies on the parabola such that $\angle NSK=60°$, is a conic whose eccentricity is 'e'. The value of $(e-2)^2$ is

18. The straight lines $2x-3y-18=0$ and $3x+2y-1=0$ intersects the circle $x^2+y^2-6x+8y+k=0$ at {A, B} and {C, D} respectively. Let P be a point on the circle such that PA.PB = 14 and PC.PD = 48 then the value of 'k' is

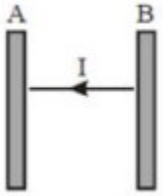
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PHYSICS **MAX.MARKS: 66**

SECTION – 1 (Maximum Marks: 18)

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For each question, choose the correct option corresponding to the correct answer.
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Negative Marks: -1 In all other cases.

19. A parallel beam of light of intensity I is incident normally on a plane surface A which absorbs 50% of the incident light. The reflected light falls on B which is perfect reflector, the light reflected by B is again partly reflected and partly absorbed and this process continues. For any light incident on A, absorption coefficient and reflection coefficient are both 0.5. The pressure experienced by A due to light is

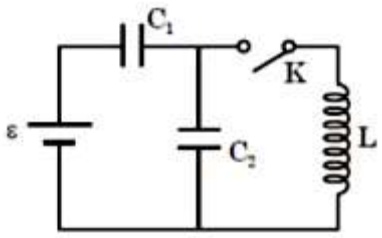


A) $\frac{1.5I}{c}$ B) $\frac{I}{c}$ C) $\frac{3I}{2c}$ D) $\frac{3I}{c}$

20. Two carnot engines A and B are operated in series. The first one A receives heat at 1200K and rejects to a reservoir at T K. The second engine B receives the heat rejected by the first engine and in turn rejects to a heat reservoir at 300 K. Calculate the value of T, when work outputs of the two engines are equal

A) 600 K B) 750 K C) 450 K D) 900 K

21. Consider the electrical circuit given below : initially chargeless capactitors C_1 and C_2 were connected to a battery and at certain moment, the key K will be closed. After the moment, current and voltage will start oscillating. For these oscillations



A) When charge on C_2 is maximum, current through C_1 is same as current in L

B) When charge on C_1 is maximum, power delivered by battery is positive

C) Charge on C_2 is maximum when current through L is maximum

D) None of these

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22. Two underdamped oscillators are known to have the same natural frequency ω_0. The mass and damping coefficient of the first oscillator are m_1 and b_1, and the mass and damping coefficient of the second oscillator are m_2 and b_2, respectively. A sinusoidal driving force of $F_{ext} = F_0 \cos \omega t$ is applied to each oscillator. Starting with ω far from ω_0 the driving force is tuned in order to observe resonant behavior. If $m_1 = 4m_2$ and $b_1 = 2b_2$, then which one of the following statements concerning the driven oscillations is **CORRECT**?

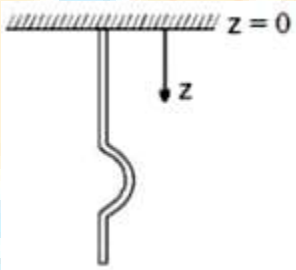
A) The resonant peak of the first driven oscillator is higher and narrower than that of the second oscillator

B) The resonant peak of the first oscillator is higher and wider than that of the second oscillator

C) The resonant peak of the first driven oscillator is lower and wider than that of the second oscillator

D) The resonant peak of the first driven oscillator is lower and narrower than that of the second oscillator

23. A rope hangs from a rigid support. A pulse is set by jiggling the bottom end. We want to design a rope in which velocity v of pulse is independent of z, the distance of the pulse from fixed end of the rope. If the rope is very long the desired function for mass per unit length $\mu(z)$ in terms of μ_0 (mass per unit length of the rope at the top ($z = 0$), g, v and z is



A) $\mu(z) = \mu_0 e^{-\left[g/v^2\right]z}$ B) $\mu(z) = \mu_0 e^{-\left[g/2v^2\right]z}$

C) $\mu(z) = \mu_0 \log_e\left(\frac{g}{v^2}\right)z$ D) $\mu(z) = \mu_0 e + \left(\frac{v^2}{g}\right)z$

24. In single slit Fraunhofer diffraction pattern, resultant amplitude at the center of central maxima is A_0 and it is A at an angular position of 30° (with respect to central axis).

Then $\frac{A}{A_0}$ will be [Slit width a = λ]

A) $\frac{\pi}{2}$ B) $\frac{2}{\pi}$ C) $\frac{\pi}{4}$ D) $\frac{4}{\pi}$

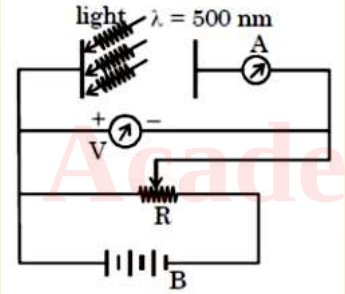
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SECTION - 2 (Maximum Marks : 24)

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25. A clean surface of a metal in a vacuum tube is irradiated with a monochromatic light of wavelength 500 nm and the resulting photoelectric current is measured by the ammeter A, as shown in the circuit. It is found that if the voltmeter (V) reading is greater than 0.8 V, the ammeter shows zero reading. Now, with a second source of light, the ammeter reading becomes zero when the voltmeter reading is gradually increased to 1.2 V. Then choose the **CORRECT** option(s), [Battery B with rheostat R makes a potential divider circuit] [Given: $h = 6.6\times10^{-34}$ Js $c = 3\times10^{8}$ ms^{-1}, $e = 1.6\times10^{-19}$ C]



A) the frequency of second source is 6.97×10^{14} Hz

B) the kinetic energy of photoelectrons due to first source is less than or equal to 0.8 eV

C) the work function of emitting surface is 1.68 eV

D) the kinetic energy of photoelectrons due to second source is less than or equal to 1.2eV

26. A partition divides a container having insulated walls into two compartments I and II. The same gas fills the two compartments whose initial parameters are given in the figure. The partition is a conducting wall which can move freely without friction. Which of the following statement(s) is/are **CORRECT**, with reference to the final equilibrium position ?



For More Join: @IITJEE_Advanced

A) The pressure in the two compartments are equal

B) Volume of compartment I is $\frac{3V}{5}$

C) Volume of compartment II is $\frac{3V}{5}$

D) Final pressure in compartment I is $\frac{5P}{3}$

27. There exists a uniform electric field $E\hat{j}$ and a uniform magnetic field $B\hat{k}$ and no gravitational field in region of space. A particle at origin of mass m, charge +q is given velocity $\vec{v}_1 = +\frac{E\hat{i}}{B}$ or $\vec{v}_2 = \frac{+\sqrt{3}E\hat{j}}{B}$ at t = 0. Then choose the **CORRECT** option(s)

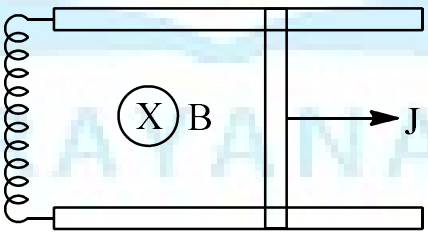
A) For particle velocity $\vec{v}_1 = +\frac{E\hat{i}}{B}$, electric force is balanced by magnetic force.

B) For particle velocity $\vec{v}_2$, particle will attain maximum y – coordinate for the first time at $\frac{2\pi m}{3qB}$

C) For particle velocity $\vec{v}_2$, maximum y – coordinate is $\frac{3mE}{qB^2}$.

D) For particle velocity $\vec{v}_2$, minimum y – coordinate is $\frac{-mE}{qB^2}$

28. Two parallel resistance less rails are connected by an inductor of inductance L at one end as shown in the figure. A magnetic field B exists in the space which is perpendicular to the plane of the rails. Now a conductor of length ℓ and mass m is placed transverse on the rails and given an impulse J towards the rightward direction. Then choose the **CORRECT** option (s)



A) Velocity of the conductor is half of the initial velocity after a displacement of the conductor $d = \sqrt{\frac{3J^2L}{4B^2\ell^2m}}$

B) Current flowing through the inductor at the instant when velocity of the conductor is half of the initial velocity is $i = \sqrt{\frac{3J^2}{4Lm}}$

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C) Velocity of the conductor is half of the initial velocity after a displacement of the conductor $d = \sqrt{\frac{3J^2L}{B^2\ell^2m}}$

D) Current flowing through the inductor at the instant when velocity of the conductor is half of the initial velocity is $i = \sqrt{\frac{3J^2}{mL}}$

29. A vessel containing water is kept on an inclined plane of coefficient of friction μ. The angle made by the free surface of liquid with horizontal as shown in steady state is θ. If surface is as shown in figure (a), mark the angle θ as positive and if it is as shown in figure (b) mark the angle θ as negative. Then choose the **CORRECT** option(s)

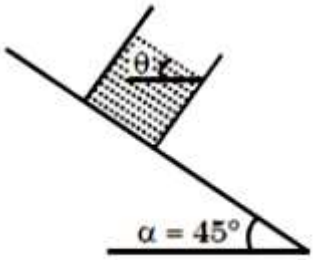


Figure (a)

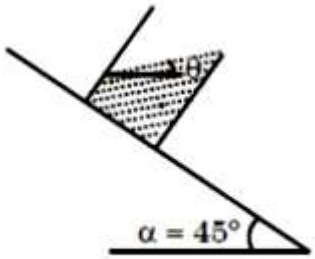


Figure (b)

A) If $\mu = 1.2$, then $\theta = 0°$

B) If $\mu = 0$, then $\theta = 45°$

C) If $\mu = \frac{3}{4}$, then $\theta = 15°$

D) If $\mu = \frac{3}{4}$, then $\theta = -15°$

30. At the equator on a spherical planet a body weighs one third of that at the pole. Radius of planet is R. The time period of revolution of the planet around its axis is equal to T = 60 min. Then choose the **CORRECT** option(s) (Take : $G = \frac{20}{3} \times 10^{-11} Nm^2/kg^2$)

A) The average density ρ of the planet is $\frac{\pi}{192} \times 10^6 kg/m^3$

B) The height of geostationary satellite from the surface of planet is $\left(\sqrt[3]{\frac{3}{2}} - 1\right)R$

C) At a latitude of $60°$, the weight of the body is half the weight at pole

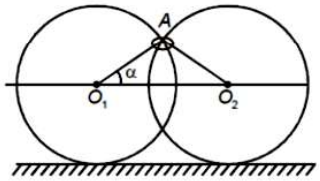
D) The geostationary satellite can be in a plane passing through the poles

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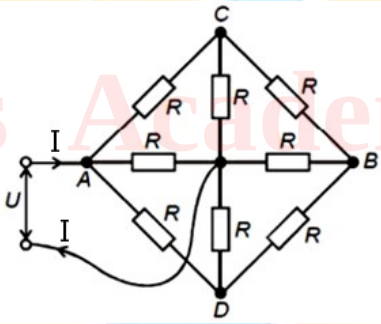
SECTION - 3 (Maximum Marks : 24)

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For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round- off** the value to **TWO** decimal places.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

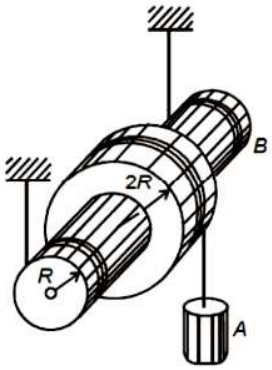
31. A ring of mass m connecting freely two identical thin hoops of mass M each starts sliding down. The hoops move apart over a rough horizontal surface. Determine the acceleration a (in m/s^2) of the ring at the initial instant if $\angle AO_1O_2 = 60°$ (Figure), neglecting the friction between the ring and the hoops (given $m = 2M$ & $g = 10\ m/s^2$) (There is no slipping between hoop and ground)



32. The resistance of each resistor in the circuit diagram shown in figure is the same and equal to R. The voltage across the terminals is U. The current I in the leads if their resistance can be neglected is given by $\frac{xU}{R}$. Find the value of 'x'.

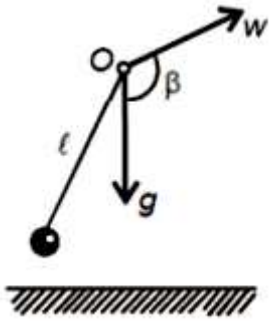


33. In the arrangement shown in figure, weight A possesses mass m, a symmetric pulley B possesses mass M. Also known are the moment of inertia I of the pulley relative to its axis and the radii of the pulley R and 2R. The mass of the threads is negligible. The acceleration of the weight A after the system is set free from rest is given by $a = \frac{3g(pM+qm)}{\left(rM+sm+\frac{I}{R^2}\right)}$. Where p,q,r,s are integers. Find the value of **p + q + r + s**.



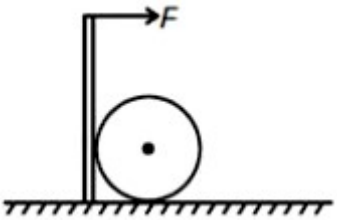
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34. Find the time period (in sec) of small oscillations of a mathematical pendulum of length ℓ if its point of suspension O moves relative to the Earth's surface in an arbitrary direction with a constant acceleration w (figure). It is given that $\ell = 21$ cm, $w = \frac{g}{2}$, and the angle between the vectors $\vec{w}$ and $\vec{g}$ equals $\beta = 120°$. (Take $g = 10\,m/s^2$)



35. In accordance with classical electrodynamics an electron moving with acceleration $\vec{w}$ loses its energy due to radiation as $\frac{dE}{dt} = -\frac{2e^2}{3c^3} w_{avg.}^2$, where e is the electron charge, c is the speed of light, $w_{avg.}$ is the average of magnitude of acceleration of the electron. The time during which the energy of an electron of mass m performing almost harmonic oscillations with frequency ω becomes $\frac{1}{\eta}$ times is given by $\frac{xmc^y}{2e^2\omega^z}\ln(\eta)$ where x, y, z are integers. Find the value of $\frac{xy}{z}$.

36. A uniform rod of mass M and length L is hinged at its lower end on a table. The rod can rotate freely in vertical plane and there is no friction at the hinge. A solid spherical ball of mass M and radius $R = \frac{L}{3}$ is placed in contact with the vertical rod and a horizontal force F is applied at the upper end of the rod. The horizontal component of hinge force acting on the rod immediately after force F starts acting is $\frac{xF}{y}$. Where x, y are co-primes. Find the value of x + y.
(Assume floor is sufficiently rough and the rod is smooth)



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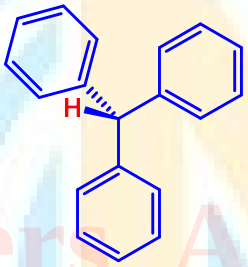
CHEMISTRY **MAX.MARKS: 66**

SECTION – 1 (Maximum Marks: 18)

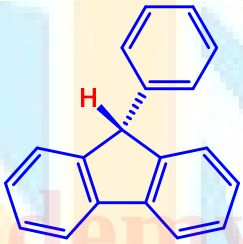
This section contains SIX (06) questions.
Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer.
For each question, choose the correct option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If only the correct option is chosen.
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)
Negative Marks: -1 In all other cases.

37. The heating of NH_4NO_2 at $60-70°C$ and NH_4NO_3 at $200-250°C$ is associated with the formation of nitrogen containing compounds X and Y , respectively. The only correct statement regarding "X" & "Y" is
A) N – N bond order in Y is less than 2
B) N – N bond order in X is 2
C) "X" and "Y" have equal number of lone pair of electrons
D) Both "X" and "Y" are diamagnetic

38. Select the **INCORRECT** option regarding the given molecules [P] & [Q]



[P]

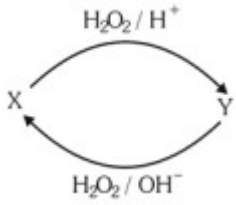


[Q]

A) Conjugate base of [Q] is aromatic, where as conjugate based of [P] is antiaromatic
B) pK_a of [P] is more than [Q]
C) Dissociation constant of [P] is less than [Q]
D) All

39. A closed vessel contains 10g of an ideal gas X at 300 K, which exerts 2 atm pressure. At the same temperature, 80 g of another ideal gas Y is added to it and the pressure becomes 6 atm. The ratio of most probable velocities of Y and X at 300 K is
A) $2\sqrt{2}:\sqrt{3}$ B) $2\sqrt{2}:1$ C) 1 : 2 D) 2 : 1

40.



$Z \xrightarrow{H_2O_2/OH^{\ominus}}$ Yellow solution

'X', 'Y' and Z respectively are :

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A) $'X' \Rightarrow K_4\left[Fe(CN)_6\right]; 'Y' \Rightarrow K_3\left[Fe(CN)_6\right]; Z = CrCl_3$

B) $'X' \Rightarrow CrCl_3; 'Y' \Rightarrow CrO_4^{2-}; Z = Cr(OH)_3$

C) $'X' \Rightarrow MnCl_2; 'Y' \Rightarrow MnO_2.xH_2O; Z = Mn(OH)_2$

D) $'X' \Rightarrow Ag; 'Y' \Rightarrow Ag_2O; Z = AgOH$

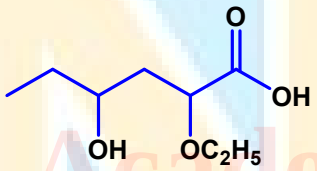
41. In the extraction of Aluminium metal, one of the process is summarized as follows :

$$\underset{\text{(bauxite Impure)}}{Al_2O_3.3H_2O} \xrightarrow[\text{conc.'A'}]{\text{hot}} \underset{\text{(aqueous solution)}}{\text{Product 'B'}} \xrightarrow[\text{to change pH}]{\text{bubble gas 'C'}} \text{Pure almunia} \xrightarrow{\text{mix with 'D' melt at 1000°C}}$$

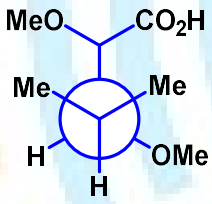
$$\xrightarrow[\text{with carbon electrodes}]{\text{electrolyse molten material}} \text{Aluminium at 'E', select the best combination for A to E.}$$

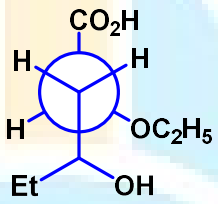
	A	B	C	D	E
A)	NaOH	Al^{3+}	HF	Na_3AlF_6	Cathode
B)	NaOH	$NaAlO_2$	CO_2	NaF	Anode
C)	H_2SO_4	$Al_2(SO_4)_3$	NH_3	Na_3AlF_6	Cathode
D)	NaOH	$NaAlO_2$	CO_2	Na_3AlF_6	Cathode

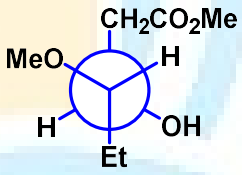
42. Which of the following Newman projection is **not** a structural isomer with given compound [**X**]



[X]

A) 

B) 

C) 

D) All

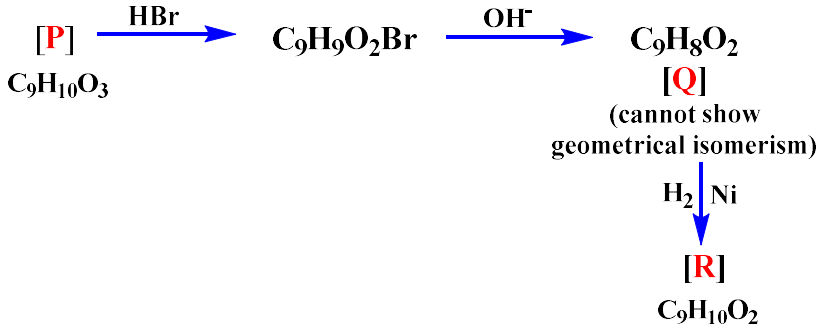
SECTION - 2 (Maximum Marks : 24)

This section contains SIX (06) questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

43. Compound [**P**] $C_9H_{10}O_3$, (can show optical isomerism) gives a positive CrO_3/H_2SO_4 test and is oxidized by hot $KMnO_4$ to benzoic acid. [**P**] is converted by the following sequence of reactions into [**R**]:

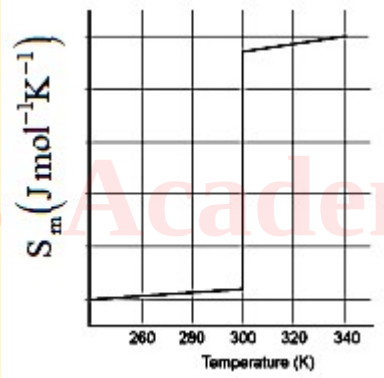
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Choose the **CORRECT** option(s)

A) [P] on heating with con H_3PO_4 gives [Q] acid
B) [R] can show optical isomerism
C) [P] can give neutral $FeCl_3$ test
D) [P] and [R] can show diastereomerism

44. The graph given below shows the molar entropy change (S_m) v/s temperature (T) for a substance undergoing solid – liquid phase transition at its normal melting point $T_m = 300K$. {Given $\Delta H_{fusion} = 6kJ/mol$} Which of the following statements are **CORRECT**.



A) The entropy change during the fusion at T_m is 20 J $mol^{-1}K^{-1}$
B) Entropy change associated with heating the solid isobarically from 250 K to 300 K

$$\Delta S = \int_{250}^{300} \frac{Cp_{(s)}}{T} dT$$

C) At 300 K substance exist in liquid state only
D) At T_m, $\Delta H = T\Delta S$ for melting of solid

45. **CORRECT** statement(s) about solubility of $AgCl(s)$ is/are ?

K_{SP} of $AgCl = 10^{-10}M^2$, K_f of $\left[Ag(NH_3)_2\right]^+_{(aq)} = 10^8$

A) Solubility in pure water is $10^{-5}M$
B) Solubility of AgCl is more in aqueous $AgNO_3$ solution than in water
C) Solubility of AgCl in 2M $NH_3(aq)$ solution is approximately 0.166 M
D) Solubility of AgCl is lesser in aqueous NaCl solution than in water

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46. Find **CORRECT** statements for coordination compounds

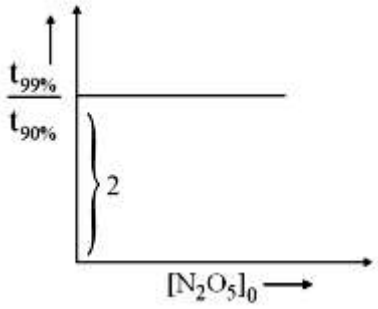
A) $\left[PtCl_2(NH_3)_2\right]$ & $\left[Cu(NH_3)_4\right]^{2+}$ are square planar complexes

B) In $\left[Fe(C_2O_4)_3\right]^{3-}$, geometrical isomerism does not exist while optical isomerism exists

C) CO bond order : $\left[Ni(CO)_4\right] > \left[Co(CO)_4\right]^- > \left[Fe(CO)_4\right]^{2-}$

D) $\left[NiCl_2(PPh_3)_2\right]$ has two geometrical isomers

47. For reaction $2N_2O_5(g) \rightarrow 4NO_2 + O_2(g)$ and $t_{90\%} = 10$ sec then select the correct options. (Use log (2) = 0.3010)



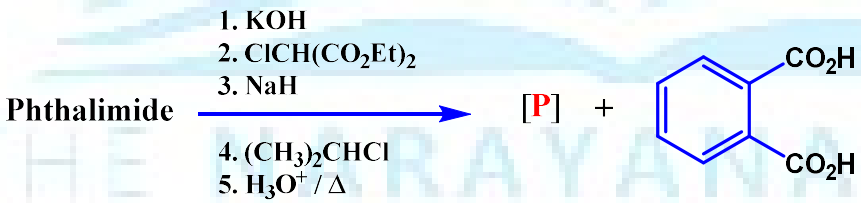
A) Reaction is of 1st order

B) Time to complete 50% reaction is 3.01 sec

C) Time to complete definite fraction of reaction is independent of initial concentration of reactant

D) $t_{15/16} = 2t_{3/4}$

48. Select **CORRECT** statement(s) about [P]



A) L isomer of [P] is essential amino acid

B) [P] can give positive carbylamine test

C) [P] on reaction with HNO_2 liberate N_2 gas

D) [P] is having two chiral centers.

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SECTION - 3 (Maximum Marks : 24)

This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**
For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round- off** the value to **TWO** decimal places.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

49. Adsorption of phenol from its aqueous solution on to fly ash obeys Freundlich isotherm. At a given temperature, from $10\,mg\,g^{-1}$ and $16\,mg\,g^{-1}$ aqueous phenol solutions, the concentrations of adsorbed phenol are measured to be $4\,mg\,g^{-1}$ and $10\,mg\,g^{-1}$, respectively. At this temperature, the concentration $\left(in\ mg\,g^{-1}\right)$ of adsorbed phenol from 32 mg g^{-1} aqueous solution of phenol will be 10^x, then 'x' is

Use : $\log_{10} 2 = 0.3$

50. Vessel – 1 contains w_2g of a non-volatile solute X dissolved in w_1 g of water. Vessel – 2 contains w_2 g of another non-volatile solute Y dissolved in w_1 g of water. Both the vessels are at the same temperature and pressure. The molar mass of X is 60% of that of Y. The van't Hoff factor for X is 20% more than that of Y for their respective concentrations.

The elevation of boiling point for solution in Vessel – 1 is ……% of the solution in Vessel – 2

51. Some reactions are given below (correct / incorrect included) for slag formation in metallurgy. Identify the value of {(X – Y) × Z} where
X = sum of "correct" reaction tags which include acidic impurity
Y = sum of "correct" reaction tags which include basic impurity
Z = sum of "incorrect" reaction tags combination(s) ?
(For example, if (1), (4) and (5) are correct then X = 1 + 4 + 5 = 10)

	Impurity		Flux	Slag
1)	CaO	+	$SiO_2 \rightarrow$	Calcium silicate
2)	FeO	+	$SiO_2 \rightarrow$	Iron silicate
3)	SiO_2	+	$CaCO_3 \rightarrow$	Calcium silicate + CO_2
4)	SiO_2	+	$MgCO_3 \rightarrow$	Magnesium silicate + CO_2
5)	P_4O_{10}	+	$SiO_2 \rightarrow$	Phospho silicate
6)	P_4O_{10}	+	$CaCO_3 \rightarrow$	Clacium phosphate + CO_2

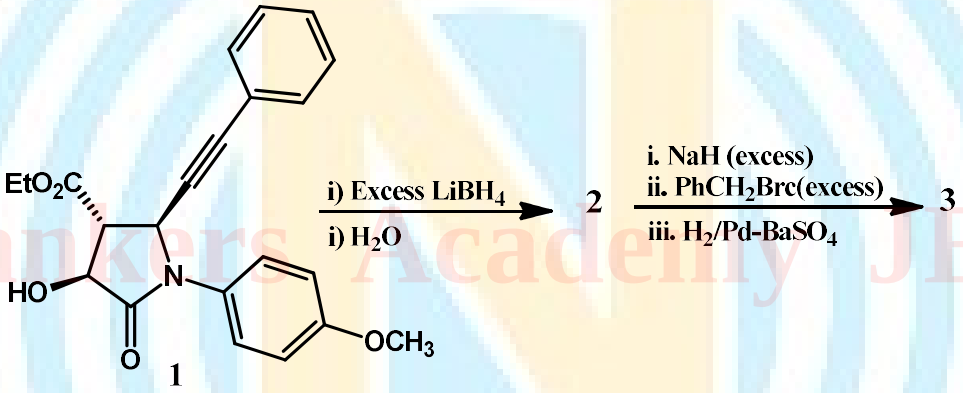
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52. During hydrolysis of ester is acidic medium $\text{ester} + H_2O \xrightarrow{H^+} \text{acid} + \text{alcohol}$ rate law is given as $r = k[\text{ester}]\left[H^+\right]$ and volume of base used at different time intervals is given as at pH = 3

t (in sec)	0	10	∞
v_b (in mL)	50	100	150

Value of rate constant at 'k' will be (in units of $M^{1-n}\,sec^{-1}$) where n = order of the reaction

53. Lithium borohydride ($LiBH_4$) is a useful reducing agent, as it is more selective than $LAlH_4$ but less selective than $NaBH_4$. Unlike the later, $LiBH_4$ will reduce esters. However, it will not reduce amides. This selectivity is frequently employed by synthetic organic chemists. In the following reaction sequence, 50.6gm of product (3) is obtained with yield of 65% and 75% respectively in 1st and 2nd steps. Find the number of moles of reactant used in this reaction? (C=12, O=16, N=14, H=1)



54. Treatment of ter-butyl benzene with CO/HCl in the presence of anhydrous $AlCl_3/CuCl$ followed by heating with CH_3CHO/dil NaOH gives compound [**P**] as a major product. Compound [**P**] up on reaction with tollens reagent followed by esterification with methyl alcohol and then hydrogenation gives [**Q**]. Reaction of [**Q**] with CH_3MgBr (excess) followed by addition of H_2O gives [**R**]. Compound [**R**] on treatment with H_2SO_4 at 0° C followed by treatment with succinic anhydride/$AlCl_3$ gives [**S**]. The reaction of compound [**S**] with Zn–Hg + Con. HCl followed by treatment with con H_3PO_4 (cat) gives [**T**]. Reaction of [**T**] with CH_3MgBr followed by heating with con H_2SO_4(cat) gives corresponding [**U**]

Number of methyl groups in [**U**] is?

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